Micropin arbiter: next steps

Reflex coils, reliable switch delivery, deadman safety, and ROMulator bring-up - 4 September 2026

Outcome

Keep the 8085 as a small, deterministic hardware arbiter. The host owns game rules, including tilt and game-over behavior. The arbiter may react autonomously only where latency and electrical safety require it: short coil timing, configured switch-to-coil reflexes, and a communications-loss deadman.

What already exists

Area	Current implementation
Host output mailbox A	$3000-$303f: sequence, 64 lamp bits, 32 display bytes, 32 coil command bits, pitch and duration.
8085 switch mailbox B	$3040-$307f: sequence/ack, 32 raw inductive samples, and ports 0, 1, 4 and 5.
Coil configuration C	$3080-$30bf: sequence/ack, 32 duration bytes, cancel-on-clear mask and renew-while-active mask.
Local image L	$2200-$223f: stable copy of A, manifested only after the complete copy.
Safety already present	All coils are finite leases. Non-renewing one-shots are the default; holdable mechanisms require explicit renewal policy.
Simulation transport	MAME localhost TCP protocol MPX1 and Python wrapper. The real Pico CDC/shared-RAM bridge is not built yet.

Settled design decisions

- The host remains the rules engine. Tilt is reported to and enforced by the host; the 8085 does not invent tilt consequences.
- All coils may be made reflex-capable, but reflex behavior is opt-in and table-driven. A disabled mapping remains host-controlled.
- Bumpers, slingshots and flippers are the primary reflex candidates. Cups and the launcher may remain host-directed where state or sequencing matters.
- A master reflex/coil permission acts as the 'everyone out of the pool' control. Clearing it cancels live timers, de-energizes every coil and blocks new reflexes.
- Host disappearance is different from game tilt: a deadman timeout independently removes coil power even if the last command requested a hold.
- The host owns lamp flashing and complex sound sequencing. The arbiter presents literal output state.

Reflex engine to add

A reflex is a configured low-latency mechanical response, not a game rule. The same switch event must still reach the host for scoring, lamps, sound and state transitions.

Component	Proposed behavior
Input normalization	Assign stable IDs to 32 inductive contacts plus cabinet inputs that are eligible for reflex use.
Mapping	Each eligible input maps to disabled or to one coil number. Coil duration continues to come from the existing per-coil duration table.
Edge/repeat policy	Momentary devices fire on an accepted edge and do not restart while their timer is live. Flippers may renew while held. Add a short configurable re-arm interval if bench testing shows mechanical bounce or insufficient return time.
Global gate	Reflex master enable must be true. A host tilt/game-over transaction clears it and all live coil timers atomically.
Host command coexistence	The effective coil state is the union of permitted reflex leases and explicit host leases, subject to the global gate and deadman.
Event publication	Firing a reflex never consumes the event. It also records the hit for reliable delivery to the host.

Recommended control precedence

```
coil output = communications_alive AND master_coil_enable AND (active_reflex_lease OR active_host_lease)
```

A host tilt response should send one coherent update: clear master enable, clear coil commands, cancel active timers, and select the scolding lamps/sound. A small normal round-trip delay is acceptable.

Deadman behavior

- Renew the deadman only when the 8085 accepts a complete, sequence-valid host transaction - not merely when bytes on the aperture change.
- On expiration: clear all 32 coil timers, write all four coil ports off, disable reflexes, and silence any sustained tone. Lamps/displays may remain at their last state for diagnosis.
- Require an explicit enable in a later valid transaction before reflexes resume. Sequence wraparound must not accidentally re-arm outputs.
- Make the deadline configurable but give ROM a conservative default. Bench-test host disconnect, process crash, USB unplug and Pico reset.

Reliable switch delivery

The present B mailbox publishes sampled levels. That is useful for bring-up, but a short bumper contact must not disappear just because the host was late. Reflex firing and event capture should occur together.

Option	Strength	Tradeoff
Latched bitmap	Small and simple	Collapses repeated hits on the same input before acknowledgment.
Per-input counters	Preserves repeated hits up to saturation	More RAM and a larger transfer; no ordering between different inputs.
Small event FIFO (preferred)	Preserves identity and order; easy to timestamp approximately	Needs producer/consumer indices and overflow behavior.

A practical first version is a fixed local FIFO of two-byte records: normalized input ID plus an 8-bit arbiter tick. Publish batches through an additional 8085-owned mailbox in the unused portion of $3000-$31ff. Retain raw B snapshots for diagnostics.

Latency budget and optimization

Path	Estimated 8085-only onset	Use
Current full host transaction	About 7-10 ms before USB/host time	Cups, lamps, displays and non-critical control.
Reordered host path	Roughly 1 ms better	Process coils before display/lamp manifestation.
Local reflex target	About 1-2 ms after accepted switch detection	Bumpers, slingshots and flippers.

The modeled RST 6.5 coil clock is 227 Hz, or about 4.4 ms per tick. A one-tick lease has phase-dependent width; two ticks are a more defensible minimum. The current safe default of five ticks is about 22 ms.

Implementation order after the ROMulator arrives

Step	Deliverable	Acceptance check
1	Scope the real expansion RAM and bus timing; confirm $3000-$31ff ownership and write visibility.	Stable bidirectional bytes under repeated reads/writes; no bus contention.
2	Add USB CDC command transport to Pico while retaining ROM mass storage, or decide on an alternate control interface.	Host can read/write aperture blocks continuously without corrupting ROM files.
3	Port the Python bridge API from TCP to the Pico transport.	Same demos run against MAME and hardware with transport selected at construction.
4	Add deadman and master-enable semantics before reflexes.	Every failure injection turns all coils off within the chosen deadline.
5	Add event capture and measure raw input-to-publication latency.	No missed synthetic or physical test pulses; overflow is visible.
6	Add one reflex mapping for a low-risk dummy load, then a bumper.	Scope verifies onset and pulse width; repeated command cannot hold a one-shot coil on.
7	Enable slingshots and flippers with explicit renewal policy.	Both flippers hold independently; release and deadman remove power promptly.
8	Connect host tilt/game-over control.	Tilt transaction cancels all mechanisms and applies host-selected sound/lamps.
9	Mirror protocol extensions in MAME and automate regression tests.	Identical Python behavior against simulator and physical bridge.

Bench-test guardrails

- Begin with logic-level observation or dummy loads, then one fused coil at a time.
- Measure actual RST cadence, switch publication delay, reflex onset, coil pulse width and host-deadman cutoff on a scope.
- Verify active-low polarity at power-up and during Pico/8085 reset. Outputs must default off before interrupts and host traffic are enabled.
- Record final per-coil safe durations and renewal/cancel policies as versioned configuration, not scattered Python constants.

Explicitly deferred

Full playfield physics, animation, AY-3-8912 support, perfect oscillator modeling, and autonomous 8085 game rules remain out of scope for the next hardware bring-up.